Executive Summary

Market-Signal-Enhanced Delta Hedging on Historical Equity Paths

Research question. Can freely available, point-in-time market signals improve replication of standardized European-call payoffs relative to fixed-volatility Black–Scholes hedging?

Objective and scope

We test whether information beyond historical SPY prices improves dynamic hedging. The experiment uses observed SPY, VIX, volume, sector-return, and short-rate data, while defining standardized European calls instead of claiming access to historical option quotes. It therefore measures payoff replication on realized market paths, not historical bid–ask execution, early exercise, or contract-level volatility-surface fit. The option claim, premium convention, rebalancing calendar, and transaction-cost model are identical across strategies.

Mathematical framework

For a call with spot S_t, strike K, time to maturity τ, risk-free rate r, and hedge volatility σ_t, every strategy holds the Black–Scholes delta

Black–Scholes delta $$\Delta_t = N(d_{1,t}), \qquad d_{1,t} = \frac{\log(S_t/K) + (r + \frac{1}{2}\sigma_t^2)\tau}{\sigma_t\sqrt{\tau}}.$$

The strategies differ only in the volatility supplied to this common engine:

- **FV-BS:** trailing 21-day realized volatility observed at initiation and then held fixed;
- **Rolling-BS:** trailing 21-day realized volatility updated on every hedge date;
- **VIX-BS:** contemporaneous VIX, used as a freely available option-market volatility proxy; and
- **MSA-Delta:** $\sigma_t^{\mathrm{MSA}} = m(Z_t)\widehat{\sigma}_t$, where $\widehat{\sigma}_t$ is rolling realized volatility and $m(Z_t)$ depends on a market-risk state.

After standardizing all features with development-period means and standard deviations, MSA-Delta uses

Risk score $$R_t = 0.30\,\mathrm{IVRank}_t + 0.20\,\mathrm{VIX}_t + 0.20\frac{\mathrm{RV}_{21,t}}{\mathrm{RV}_{63,t}} + 0.10\,\mathrm{VolumeShock}_t + 0.10\,\mathrm{Drawdown}_{63,t} + 0.05\,\mathrm{SectorDiv}_t + 0.05|\Delta\mathrm{Rate}_{21,t}|.$$

Training-score quantiles determine the state and multiplier:

Risk-state rule $$Z_t = \begin{cases} \text{normal}, & R_t < q_{70}, \\ \text{elevated}, & q_{70} \le R_t < q_{90}, \\ \text{stress}, & R_t \ge q_{90}, \end{cases} \qquad m(Z_t) \in \{1.00, 1.15, 1.40\}.$$

Empirical method

The split is strictly chronological. Training (April 2016–August 2020) estimates feature scaling and risk thresholds. Validation (August 2020–May 2022) selects among three predeclared MSA configurations using mean absolute hedging error (MAE). The selected moderate rule is refit on training plus validation data, and the held-out test (May 2022–December 2024) is evaluated once. Every feature at date t uses information available at or before t.

The test grid crosses maturities of 10, 21, and 42 trading days with strike/spot ratios of 0.95, 1.00, and 1.05. Episodes begin every five trading days, yielding 1,149 claims per strategy. Hedges rebalance daily and pay 5 basis points of proportional costs. Each strategy receives the same FV-BS premium within an episode. Terminal replication error is

Terminal error $E_T = V_T - \max(S_T - K, 0)$, where V_T is the hedge portfolio after stock liquidation and transaction costs.

We report MAE, root mean squared error (RMSE), loss Expected Shortfall (ES), turnover, and moving date-block bootstrap confidence intervals for paired MAE improvements. Overlapping starts expand coverage; block resampling addresses their time dependence.

Held-out results

Strategy	MAE	RMSE	95% loss ES	Avg. cost
FV-BS	**1.5523**	2.5422	6.7463	0.4631
Rolling-BS	1.6074	2.4385	6.4496	0.4690
VIX-BS	1.6956	**2.3993**	**5.5171**	**0.4580**
MSA-Delta	1.6906	2.5368	6.5819	0.4662

FV-BS has the lowest MAE, so the test does not support a blanket claim that additional signals improve average hedging accuracy. VIX-BS has the lowest RMSE, 95% loss ES, average cost, and turnover. The difference is economically meaningful: forward-looking volatility reduces some large replication errors even though it is less accurate under the average absolute-error criterion.

Uncertainty and robustness

Define paired improvement as $I = |E_{\mathrm{FV}}| - |E_{\mathrm{strategy}}|$, so $I > 0$ favors the alternative. Moving-block bootstrap estimates are -0.138 for MSA-Delta (95% CI $[-0.254, -0.039]$), -0.057 for Rolling-BS ($[-0.119, 0.006]$), and -0.148 for VIX-BS ($[-0.255, -0.033]$). Thus neither VIX-BS nor MSA-Delta improves MAE relative to FV-BS with statistical support. The central ranking also survives transaction-cost assumptions of 0, 5, and 10 basis points: FV-BS retains the lowest MAE and VIX-BS the lowest RMSE. Results are additionally reported for all nine maturity–moneyness combinations.

Conclusion and limitations

Main finding. Forward-looking option-market information improves tail-risk control, but additional composite-signal complexity does not automatically produce a better hedge.

Fixed realized volatility is best for minimizing average absolute replication error in this sample. VIX is the strongest simple benchmark for squared error and adverse-tail performance. The validation-selected MSA rule does not add robust value beyond the simpler alternatives. This negative result is useful: signal complexity must be validated out of sample rather than justified only by economic intuition.

VIX is an index-level, approximately 30-day volatility measure rather than contract-level SPY implied volatility. The controlled European claims omit historical option spreads, dividends, American early exercise, and volatility-surface effects. Block-bootstrap inference mitigates, but does not eliminate, dependence among overlapping episodes. The natural extension is to repeat the same leakage-controlled experiment using historical contract-level option quotes when licensed data become available.